

Code-Layout, Readability and Reusability

Date	20 June 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used through out the code	Yes	Proper indentation is maintained across all Python, HTML, CSS, and JavaScript files.
2	Proper File Structure	Files and folders are logically organized	Yes	Project is organized into templates, static, model, dataset, and application files.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Descriptive variable names such as life_expectancy, gni_per_capita, and predicted_hdi_score are used.
4	Function / Method Names	Functions are descriptively named	Yes	Functions like predict_hdi(), load_model(), and validate_input() clearly describe their purpose.
5	Code Comments	Inline and block comments explain logic	Yes	Important sections and functions are documented with comments for better understanding.
6	Modular Design	Code is split into reusable functions/modules	Yes	Separate modules are used for prediction, preprocessing, authentication, and report generation.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Common functionality is reused through functions to avoid duplication.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Input validation and exception handling are implemented to manage runtime errors.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Authentication Module	Python, Flask	Login and Registration	High
2	HDI Prediction Module	Python, Scikit-learn	Prediction Process	High
3	Data Preprocessing Module	Python, Pandas, NumPy	Data Cleaning and Prediction	High
4	Dashboard Visualization Module	HTML, CSS, Bootstrap, JavaScript	Prediction Dashboard	High
5	Database Module	SQLite	User Management and Prediction History	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate folders for templates, static files, models, datasets, and application logic.
Readability	5	Meaningful variable names, consistent formatting, and clear function names make the code easy to understand.
Reusability	5	Modular design with reusable components for authentication, prediction, preprocessing, and report generation.
Documentation / Comments	4	Most important functions and logic are documented with comments; additional inline comments can further improve maintainability.
Overall Score	4.8	The codebase is well-structured, maintainable, readable, and reusable, making it suitable for future enhancements and deployment.